

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Polar Coordinates and Parametric Equations

Week 4 Day 1 AM



OUTLINE

① Polar Coordinates

② Graphs of Polar Equations

③ Plane Curves and Parametric Equations

④ Summary

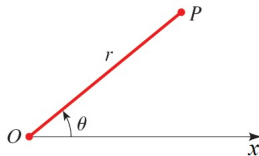
DEFINITION OF POLAR COORDINATES

Recall that the Cartesian coordinate system uses the horizontal and vertical displacements from the origin O to specify the location of a point in the Cartesian plane, but this is not the only way.

The **polar coordinate system** is a different coordinate system that uses distances and directions to specify the location of a point in the Cartesian plane. For any point P on the plane, we assign the **polar coordinates** $P(r, \theta)$, where

$r \geq 0$ is the distance from O to P

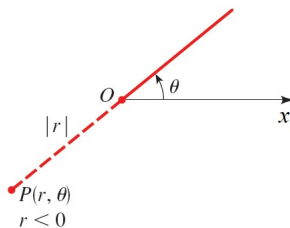
θ is the angle between the x -axis and the ray OP .



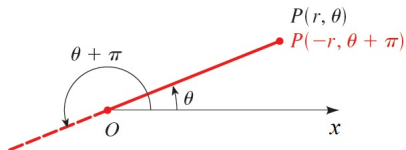
We use the convention that θ is **positive** if measured in an **anti-clockwise** direction from the positive x -axis or **negative** if measured in a **clockwise** direction.

DEFINITION OF POLAR COORDINATES

If r is negative, then $P(r, \theta)$ is defined as follows:

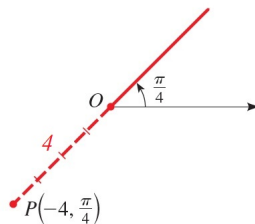
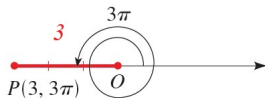
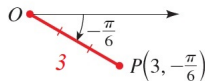
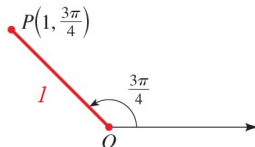


With this definition, we see that (r, θ) and $(-r, \theta + \pi)$ specify the same point.



POLAR COORDINATES - EXAMPLE

E.g.:



POLAR AND CARTESIAN COORDINATES

By using trigonometry, we can convert polar coordinates to Cartesian coordinates and vice-versa.

RELATIONSHIP BETWEEN POLAR AND RECTANGULAR COORDINATES

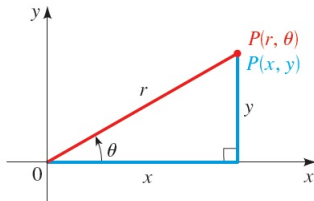
1. To change from polar to rectangular coordinates, use the formulas

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta$$

2. To change from rectangular to polar coordinates, use the formulas

$$r^2 = x^2 + y^2 \quad \text{and} \quad \tan \theta = \frac{y}{x} \quad (x \neq 0)$$

Remark: When solving for r and θ , we normally restrict them to $r \geq 0$ and $\theta \in [0, 2\pi)$ or $[\pi, -\pi)$.



POLAR AND CARTESIAN COORDINATES - EXAMPLE

E.g.: The Cartesian coordinates for the point that has polar coordinates $\left(4, \frac{2\pi}{3}\right)$ are

$$x = 4 \cos \frac{2\pi}{3} = 4 \left(-\frac{1}{2}\right) = -2, \quad y = 4 \sin \frac{2\pi}{3} = 4 \left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}.$$

E.g.: The polar coordinates for the point that has Cartesian coordinates $(-2, 2)$ are

$$r = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}, \quad \tan \theta = \frac{2}{-2} = -1$$
$$\Rightarrow \quad \theta = \frac{3\pi}{4}, \left(\neq \arctan(-1) = -\frac{\pi}{4} \right)$$

as the point $P(-2, 2)$ is in the 2nd quadrant.

ACTIVITY 1

(a) Find the Cartesian coordinates of the following points whose polar coordinates are given.

(i) $(4, \frac{\pi}{6})$

(ii) $(\sqrt{2}, -\frac{\pi}{4})$

(iii) $(5, 5\pi)$

(iv) $(6\sqrt{2}, \frac{11\pi}{6})$

(b) Find the polar coordinates ($r \geq 0$, $\theta \in [0, 2\pi)$) of the following points whose Cartesian coordinates are given.

(i) $(-1, -1)$

(ii) $(3, -4)$

(iii) $(-6, 0)$

(iv) $(0, 7)$

ACTIVITY 1: SOLUTION

$$(a) \quad (i) \quad (x, y) = \left(4 \cos \frac{\pi}{6}, 4 \sin \frac{\pi}{6}\right) = \left(4 \cdot \frac{\sqrt{3}}{2}, 4 \cdot \frac{1}{2}\right) = (2\sqrt{3}, 2).$$

$$(ii) \quad (x, y) = \left(\sqrt{2} \cos \left(-\frac{\pi}{4}\right), \sqrt{2} \sin \left(-\frac{\pi}{4}\right)\right) = \\ \left(\sqrt{2} \cdot \frac{1}{\sqrt{2}}, \sqrt{2} \cdot \left(-\frac{1}{\sqrt{2}}\right)\right) = (1, -1).$$

$$(iii) \quad (x, y) = (5 \cos 5\pi, 5 \sin 5\pi) = (5 \cos \pi, 5 \sin \pi) \\ = (5 \cdot (-1), 5 \cdot 0) = (-5, 0).$$

$$(iv) \quad (x, y) = \left(6\sqrt{2} \cos \frac{11\pi}{6}, 6\sqrt{2} \sin \frac{11\pi}{6}\right) \\ = \left(6\sqrt{2} \cos \left(-\frac{\pi}{6}\right), 6\sqrt{2} \sin \left(-\frac{\pi}{6}\right)\right) \\ = \left(6\sqrt{2} \cdot \frac{\sqrt{3}}{2}, 6\sqrt{2} \left(-\frac{1}{2}\right)\right) = (3\sqrt{6}, -3\sqrt{2}).$$

ACTIVITY 1: SOLUTION

- (b) (i) Since $(-1, -1)$ is in the 3rd quadrant and $\tan \theta = \frac{-1}{-1} = 1$, we have $\theta = \frac{5\pi}{4}$.
 $r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$. So $(r, \theta) = \left(\sqrt{2}, \frac{5\pi}{4}\right)$.
- (ii) Since $(3, -4)$ is in the 4th quadrant and $\arctan\left(-\frac{4}{3}\right) = -0.9273$ rad, so $r = \sqrt{3^2 + (-4)^2} = 5$ and $\theta = 2\pi - 0.9273 = 2\pi + \arctan\left(-\frac{4}{3}\right)$.
 $(r, \theta) = \left(5, 2\pi + \arctan\left(-\frac{4}{3}\right)\right) = (5, 5.3559)$.
- (iii) Since $(-6, 0)$ is on the negative side of the x -axis, $\theta = \pi$ and $r = \sqrt{(-6)^2 + 0^2} = 6$. So $(r, \theta) = (6, \pi)$.
- (iv) Since $(0, 7)$ is on the positive side of the y -axis, $\theta = \frac{\pi}{2}$ and $r = \sqrt{0^2 + 7^2} = 7$. So $(r, \theta) = \left(7, \frac{\pi}{2}\right)$.

CARTESIAN EQUATIONS TO POLAR EQUATIONS

The formulas in slide 6 can also be used to convert equations in Cartesian coordinates to equations in polar coordinates and vice-versa.

E.g.: We convert the parabola $x^2 = 4y$ to a polar equation by substituting $x = r \cos \theta$ and $y = r \sin \theta$:

$$x^2 = 4y \quad \Leftrightarrow \quad (r \cos \theta)^2 = 4(r \sin \theta)$$

$$\Leftrightarrow \quad r^2 \cos^2 \theta = 4r \sin \theta \quad \Rightarrow \quad r = \frac{4 \sin \theta}{\cos^2 \theta}$$

$$\Leftrightarrow \quad r = 4 \frac{\sin \theta}{\cos \theta} \frac{1}{\cos \theta} \quad \Rightarrow \quad r = 4 \tan \theta \sec \theta.$$

POLAR EQUATIONS TO CARTESIAN EQUATIONS

E.g.: We convert the polar equation $r = 5 \sec \theta$ to a Cartesian equation by substituting $r \cos \theta = x$:

$$\begin{aligned} r = 5 \sec \theta & \Leftrightarrow r = \frac{5}{\cos \theta} \\ \Leftrightarrow r \cos \theta = 5 & \Rightarrow x = 5. \end{aligned}$$

So $r = 5 \sec \theta$ is actually the vertical line $x = 5$.

E.g.: We convert the polar equation $r = 2 \sin \theta$ to a Cartesian equation by first multiplying both sides by r , followed by substituting $r \sin \theta = y$ and $r^2 = x^2 + y^2$:

$$\begin{aligned} r = 2 \sin \theta & \Rightarrow r^2 = 2 r \sin \theta \\ \Leftrightarrow x^2 + y^2 = 2y & \Leftrightarrow x^2 + y^2 - 2y = 0 \\ \Leftrightarrow x^2 + y^2 - 2y + 1 = 1 & \Leftrightarrow x^2 + (y - 1)^2 = 1. \end{aligned}$$

So $r = 2 \sin \theta$ is actually the unit circle centered at $(0, 1)$.

ACTIVITY 2

(a) Convert the following equations to polar form.

(i) $x = y$

(ii) $x^2 + y^2 = 9$

(iii) $x^2 - y^2 = 1$

(iv) $y = 3$

(b) Convert the following polar equations to Cartesian form.

(i) $r \cos \theta = 6$

(ii) $r = 4 \sin \theta$

(iii) $r^2 = \tan \theta$

(iv) $r = 1 + \cos \theta$

ACTIVITY 2: SOLUTION

(a) (i)

$$\begin{aligned} x = y & \Leftrightarrow r \cos \theta = r \sin \theta \\ \Rightarrow \cos \theta = \sin \theta & \Rightarrow \tan \theta = 1. \end{aligned}$$

(ii)

$$\begin{aligned} x^2 + y^2 = 9 & \Leftrightarrow r^2 = 9 \\ \Rightarrow r = 3. \end{aligned}$$

(iii)

$$\begin{aligned} x^2 - y^2 = 1 & \Leftrightarrow r^2 \cos^2 \theta - r^2 \sin^2 \theta = 1 \\ \Leftrightarrow r^2 (\cos^2 \theta - \sin^2 \theta) = 1 & \Rightarrow r^2 = \frac{1}{\cos^2 \theta - \sin^2 \theta} \\ \Leftrightarrow r^2 = \frac{1}{\cos(2\theta)} & \Leftrightarrow r^2 = \sec(2\theta). \end{aligned}$$

$$(iv) \quad y = 3 \Leftrightarrow r \sin \theta = 3 \Rightarrow r = 3 \csc \theta.$$

ACTIVITY 2: SOLUTION

(b) (i) $r \cos \theta = 6 \Leftrightarrow x = 6$. So $r \cos \theta = 6$ is the vertical line $x = 6$.

(ii)

$$\begin{aligned} r = 4 \sin \theta &\Rightarrow r^2 = 4 r \sin \theta \\ \Leftrightarrow x^2 + y^2 &= 4y &\Leftrightarrow x^2 + y^2 - 4y = 0 \\ \Leftrightarrow x^2 + y^2 - 4y + 4 &= 4 &\Leftrightarrow x^2 + (y - 2)^2 = 2^2. \end{aligned}$$

So $r = 4 \sin \theta$ is the circle centered at $(0, 2)$ with radius 2.

(iii)

$$\begin{aligned} r^2 &= \tan \theta &\Rightarrow x^2 + y^2 &= \frac{y}{x} \\ \Rightarrow x^3 + xy^2 - y &= 0. \end{aligned}$$

ACTIVITY 2: SOLUTION

(b) (iv)

$$r = 1 + \cos \theta$$

$$\Rightarrow r^2 = r + r \cos \theta$$

$$\Leftrightarrow x^2 + y^2 = \sqrt{x^2 + y^2} + x$$

$$\Leftrightarrow x^2 + y^2 - x = \sqrt{x^2 + y^2}$$

$$\Rightarrow (x^2 + y^2 - x)^2 = x^2 + y^2.$$

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GRAPHING POLAR EQUATIONS

Polar coordinate system has the advantage of expressing circular curves in simpler equations compared to Cartesian coordinate system. For instance, the example on slide 12:

$$r = 2 \sin \theta \quad \text{v.s.} \quad x^2 + (y - 1)^2 = 1.$$

This will be useful for evaluating some (double) integrals. Also, many curves that arise in mathematics and its applications are more easily and naturally represented by polar equations than by Cartesian equations.

On the other hand, given a general polar equation, how can we sketch its graph? One can of course convert it into the Cartesian form and then sketch the graph, for example the equations above. However, there are cases where the resulting Cartesian equation is complicated and not illuminating, for instance, the last equation in the previous activity. Thus, we need to know how to sketch the graph of an equation of the original polar form.

GRAPHING POLAR EQUATIONS

The idea behind sketching polar equations $r = g(\theta)$ is similar to the one of Cartesian equations $y = f(x)$: while we increase the independent variable, we vary the dependent variable accordingly as given by the equations.

So to graph $y = f(x)$, while we increase the x , i.e. move to the right horizontally, we vary y accordingly, i.e. we move vertically according to $y = f(x)$.

With the same principle, to graph $r = g(\theta)$, while we increase the θ , i.e. rotate around the origin anti-clockwise, we vary r accordingly, i.e. we change the radial distance from the origin according to $r = g(\theta)$. This [Geogebra applet](#) can help you to visualize this process.

ACTIVITY 3

Try to visualize the following curves before using the [Geogebra applet](#) to sketch them.

(a) $r = 3$

(b) $r = \theta$

(c) $r = 2 \sin \theta$

(d) $r = 2 + 2 \cos \theta$

(e) $r = 1 + 2 \cos \theta$

(f) $r = \cos 2\theta$

(g) $r^2 = \sin 2\theta$

(h) $r = \theta \sin \theta$

(i) $r = \sin \theta + \sin^3 \left(\frac{5\theta}{2} \right)$

(j) Create your own interesting polar curve!

COMMON POLAR CURVES

SOME COMMON POLAR CURVES

Circles and Spiral



$r = a$
circle



$r = a \sin \theta$
circle



$r = a \cos \theta$
circle



$r = a\theta$
spiral

Limaçons

$$r = a \pm b \sin \theta$$

$$r = a \pm b \cos \theta$$

$$(a > 0, b > 0)$$

Orientation depends on
the trigonometric function
(sine or cosine) and the sign of b .



$a < b$
limaçon with
inner loop



$a = b$
cardioid



$a > b$
dimpled limaçon



$a \geq 2b$
convex limaçon

Roses

$$r = a \sin n\theta$$

$$r = a \cos n\theta$$

n -leaved if n is odd

$2n$ -leaved if n is even



$r = a \cos 2\theta$
4-leaved rose



$r = a \cos 3\theta$
3-leaved rose



$r = a \cos 4\theta$
8-leaved rose



$r = a \cos 5\theta$
5-leaved rose

Lemniscates

Figure-eight-shaped
curves



$r^2 = a^2 \sin 2\theta$
lemniscate



$r^2 = a^2 \cos 2\theta$
lemniscate

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PARAMETRIC EQUATIONS

So far, we have described curves on the Cartesian plane by an equation (in rectangular or polar coordinates) that relates the two coordinates. But not all curves in the plane can be described in this way. In this section we study parametric equations, which are a general method for describing any curve.

Think of a curve as the path of a point moving in the plane; the x - and y -coordinates of the point are then functions of time. This idea leads to the following definition.

DEFINITION:

If f, g are functions defined on an interval I , then the set of points $(f(t), g(t))$ is a **plane curve** and the equations

$$x = f(t), \quad y = g(t) \quad \text{for } t \in I$$

are **parametric equations** for the curve, with **parameter** t .

⚠ It's intuitive to think of t as the time, but it's not always the case. Sometimes the parameter t can be negative or has other physical meaning.

PARAMETRIC EQUATIONS - EXAMPLE

You can find a few examples of curve described by parametric equations in this [Geogebra applet](#). The i -component is the x -component, the j -component is the y -component.

E.g.: By using this [Geogebra applet](#) to sketch the curve defined by the parametric equations

$$x = t^2 - 3t \quad y = t - 1,$$

we see that it's a parabola. If we replace t by $-t$, we obtain the new parametric equations

$$x = t^2 + 3t \quad y = -t - 1.$$

We use the same applet to sketch the curve and observe that it's the same parabola, but traced out in the opposite direction. If we substitute t by $2t$ in the original equations, we get

$$x = 4t^2 - 6t \quad y = 2t - 1.$$

The curve is again the same but it's traced out "twice as fast".

PARAMETRIZATION

The last example shows that a **parametrization** (defining a curve using parametric equations) contains more information than just the shape of the curve; it also indicates how the curve is being traced out. Parametric equations are useful in modelling motions and can be used to compute related velocity, acceleration, work done, etc. You will learn this in multivariable calculus in term 2.

E.g.: Suppose a particle is moving along the curve $y = f(x)$ from left to right, starting at the position $(x_0, f(x_0))$, then one easy parametrization is

$$x = t \quad y = f(t)$$

where t starts at x_0 . If the particle is moving from right to left, then we can parametrize its motion as

$$x = -t \quad y = f(-t)$$

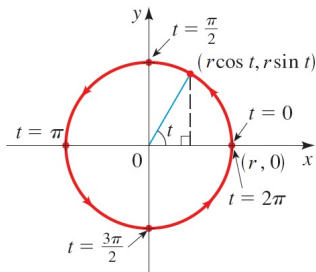
where t starts at $-x_0$. Try this for $y = x^3 + 2x^2 - x - 1$ using this [Geogebra applet](#).

EXAMPLE - MODELLING CIRCULAR MOTION

To model an anti-clockwise circular motion of radius r , we can use the following parametric equations:

$$x = r \cos t \quad y = r \sin t \quad \text{for } t \geq 0.$$

The formulas are obtained using trigonometry and polar coordinates. The parameter t here is the angle at the origin subtended by the arc from $(r, 0)$ to the point $(r \cos t, r \sin t)$:



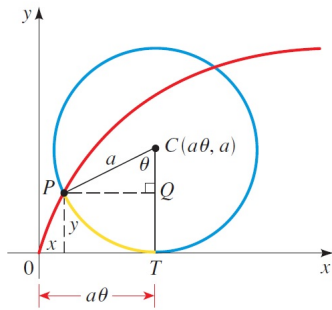
Use this [Geogebra applet](#) to visualize it.

Qn: What if the circular motion is clockwise?

EXAMPLE - THE CYCLOID

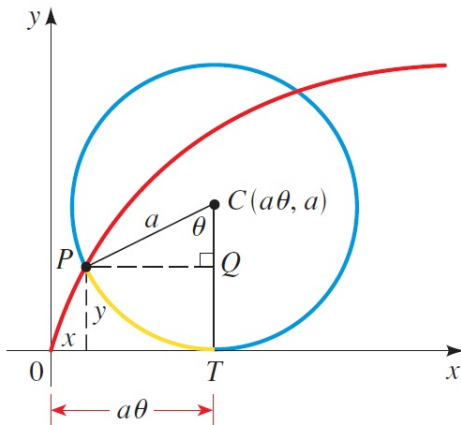
As a circle of radius a rolls along a horizontal straight line without slipping, the curve traced out by a fixed point P on the circumference of the circle is called a **cycloid**. Use this [Geogebra applet](#) to visualize it.

To model a cycloid, we let θ (in radians) be the angle subtended by the arc PT , where T is the point of contact of the circle with the x -axis, i.e. the circle has rolled a distance of $a\theta$ and hence the coordinates of the center C is $(a\theta, a)$:

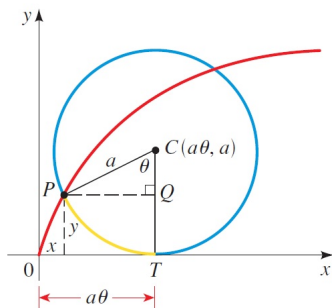


ACTIVITY 4

With the help of the figure below and trigonometry, derive parametric equations for the cycloid with θ as the parameter. Graph and verify your answer using this [Geogebra applet](#).



ACTIVITY 4: SOLUTION



$$x = d(O, T) - d(P, Q) = a\theta - a \sin \theta = a(\theta - \sin \theta),$$

$$y = d(T, C) - d(Q, C) = a - a \cos \theta = a(1 - \cos \theta),$$

for $\theta \geq 0$. The cycloid has a number of **interesting physical properties**. It is the “**curve of quickest descent under uniform gravity**”.

SKETCHING A PARAMETRIC CURVE

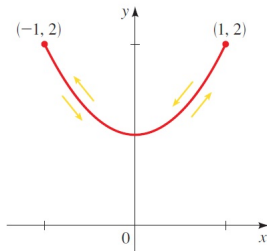
In general it's hard to sketch a parametric curve, but if we can find its equation that relates x and y by eliminating the parameter t , then it's reduced to graphing $y = f(x)$.

E.g.: Suppose we are given the parametric equations

$$x = \sin t \quad y = 2 - \cos^2 t.$$

Then

$$\begin{aligned} y &= 2 - \cos^2 t \\ &= 2 - (1 - \sin^2 t) \\ &= 1 + \sin^2 t \\ &= 1 + x^2. \end{aligned}$$



Since $x = \sin t \in [-1, 1]$, so the parametric curve is the segment of the parabola $y = 1 + x^2$ over the interval $[-1, 1]$. As $t \rightarrow \infty$, the point just oscillates back and forth on this segment.

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SUMMARY

- Polar Coordinates
 - polar coordinate system.
 - relationship between polar and Cartesian coordinates.
 - Cartesian equations to polar equations and vice-versa.
- Graphs of Polar Equations
 - graphing polar equations.
 - common polar curves.
- Plane Curves and Parametric Equations
 - parametric equations and parametrization.
 - modelling circular motion and the cycloid.
 - sketching a parametric curve.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 8.1, 8.2, 8.4. Section 8.3 is not covered in this course, but it's good to know.